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Experiment 1 – Design Star and Snowflake Schema

Aim

To design Two Star Schemas and a Snowflake Schema for a Video Streaming Platform (OTT / content-streaming analytics) using a suitable fact table and dimension tables.

Objective

- To understand the concept of Data Warehouse Schema.
- To design a Star Schema.
- To design a Snowflake Schema.

Theory

→ What is Data Warehouse Schema ?

Data warehouse schema is the logical blueprint that shows how data is organized, connected, and used for analysis in a data warehouse. It defines the tables, keys, relationships, and rules that help BI tools and analysts query data consistently.

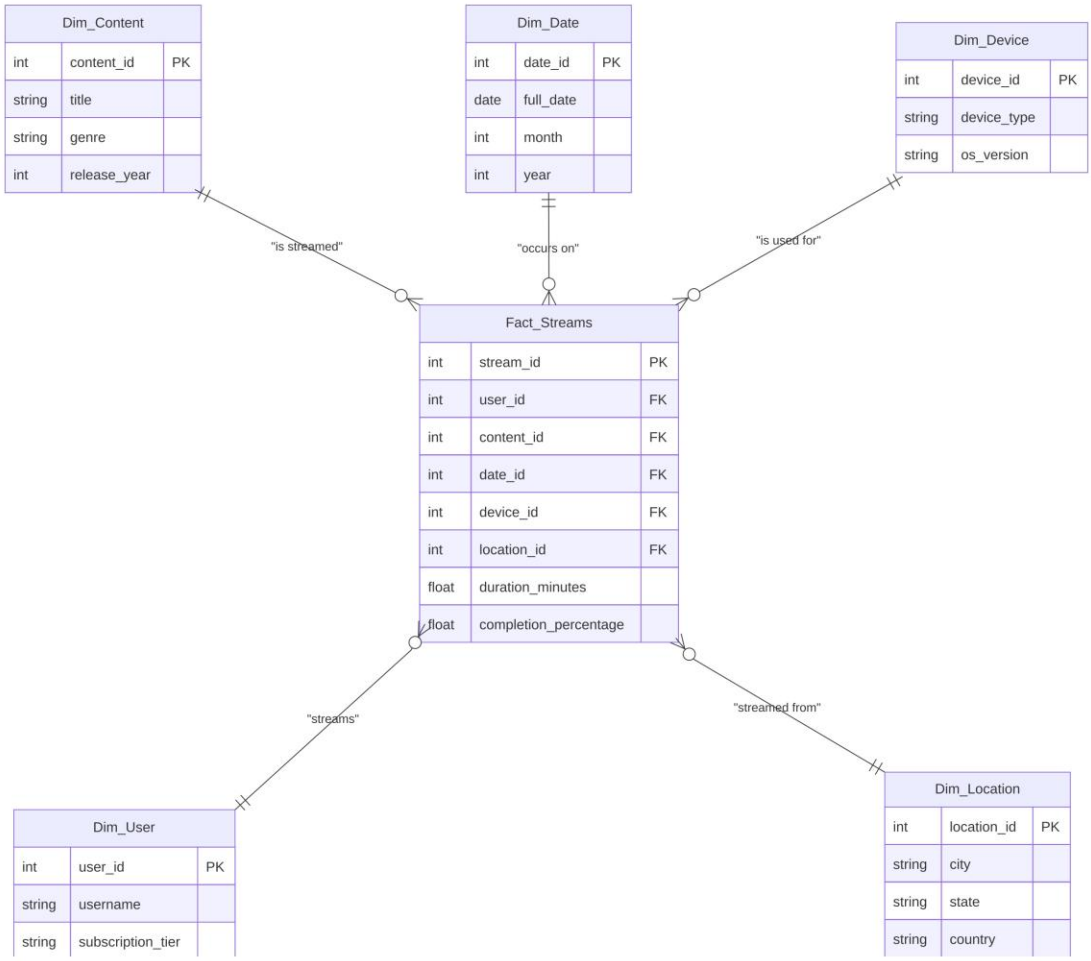
What it includes :

- Fact tables: store measurable business events, like streams, purchases, or subscriptions.
- Dimension tables: store descriptive details, like user, content, or date.
- Relationships: connect facts and dimensions so analytical queries work correctly.

Star Schema

A star schema is a way of organizing data in a data warehouse for reporting and analytics. It is called a star schema because the structure looks like a star: a central fact table is connected to several surrounding dimension tables.

Structure of Star Schema (Streaming Activity) :



Fact Table :

Fact_Streams

Column	Description
stream_id	Primary Key
user_id	Foreign Key
content_id	Foreign Key
date_id	Foreign Key
device_id	Foreign Key
location_id	Foreign Key
duration_minutes	Total minutes watched in the session
completion_percentage	Percentage of content completed

Dimension Tables

Dim_User

Column	Description
user_id	Primary Key
username	Display name of the subscriber
subscription_tier	Free / Basic / Premium plan

Dim_Content

Column	Description
content_id	Primary Key
title	Title of the movie / show / episode
genre	Genre of the content
release_year	Year the content was released

Dim_Date

Column	Description
date_id	Primary Key
full_date	Date of the stream
month	Month
year	Year

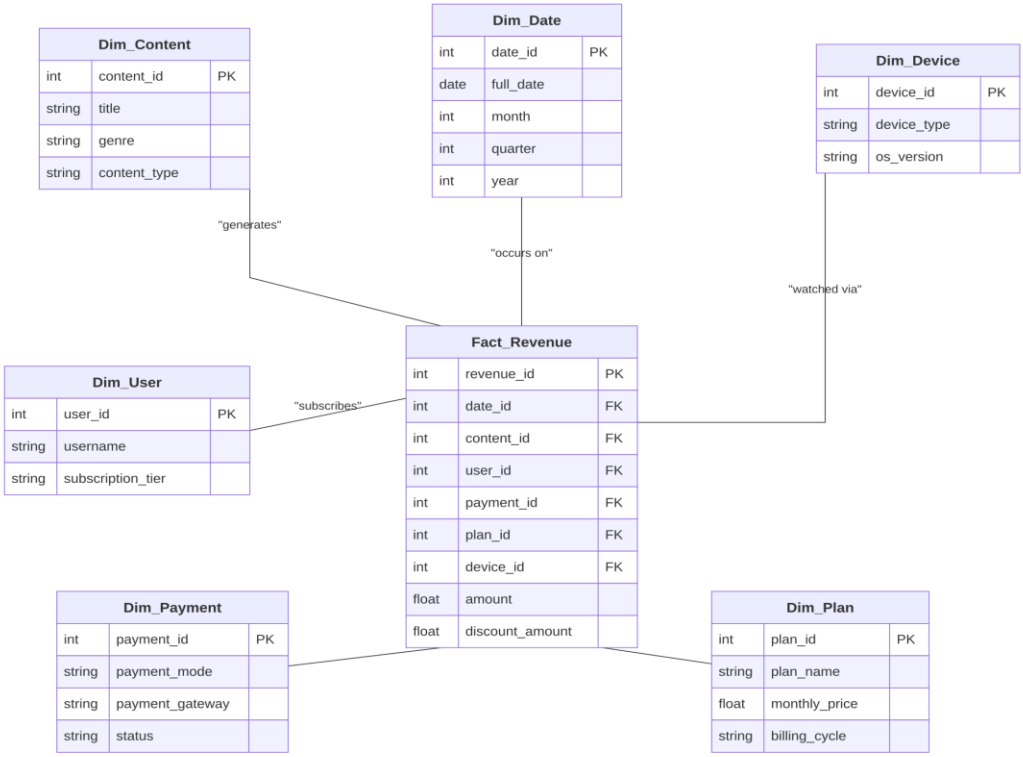
Dim_Device

Column	Description
device_id	Primary Key
device_type	Mobile / Smart TV / Laptop / Tablet
os_version	Operating system version of the device

Dim_Location

Column	Description
location_id	Primary Key
city	City the stream originated from
state	State
country	Country

Structure of Star Schema (Revenue)



Fact Table:

Fact_Revenue

Column	Description
revenue_id	Primary Key
date_id	Foreign Key
content_id	Foreign Key
user_id	Foreign Key
payment_id	Foreign Key
plan_id	Foreign Key
device_id	Foreign Key
amount	Revenue amount for the transaction
discount_amount	Discount applied on the transaction

Dimension Tables :

Dim_Date

Column	Description
date_id	Primary Key
full_date	Date of transaction
month	Month
quarter	Quarter
year	Year

Dim_Content

Column	Description
content_id	Primary Key
title	Title of the content generating revenue
genre	Genre of the content
content_type	Movie / Series / Documentary

Dim_User

Column	Description
user_id	Primary Key
username	Subscriber name
subscription_tier	Free / Basic / Premium plan

Dim_Payment

Column	Description
payment_id	Primary Key
payment_mode	Card / UPI / Wallet / Netbanking
payment_gateway	Gateway used for the transaction
status	Success / Failed / Refunded

Dim_Plan

Column	Description
plan_id	Primary Key
plan_name	Name of the subscription plan
monthly_price	Price of the plan per month
billing_cycle	Monthly / Quarterly / Annual

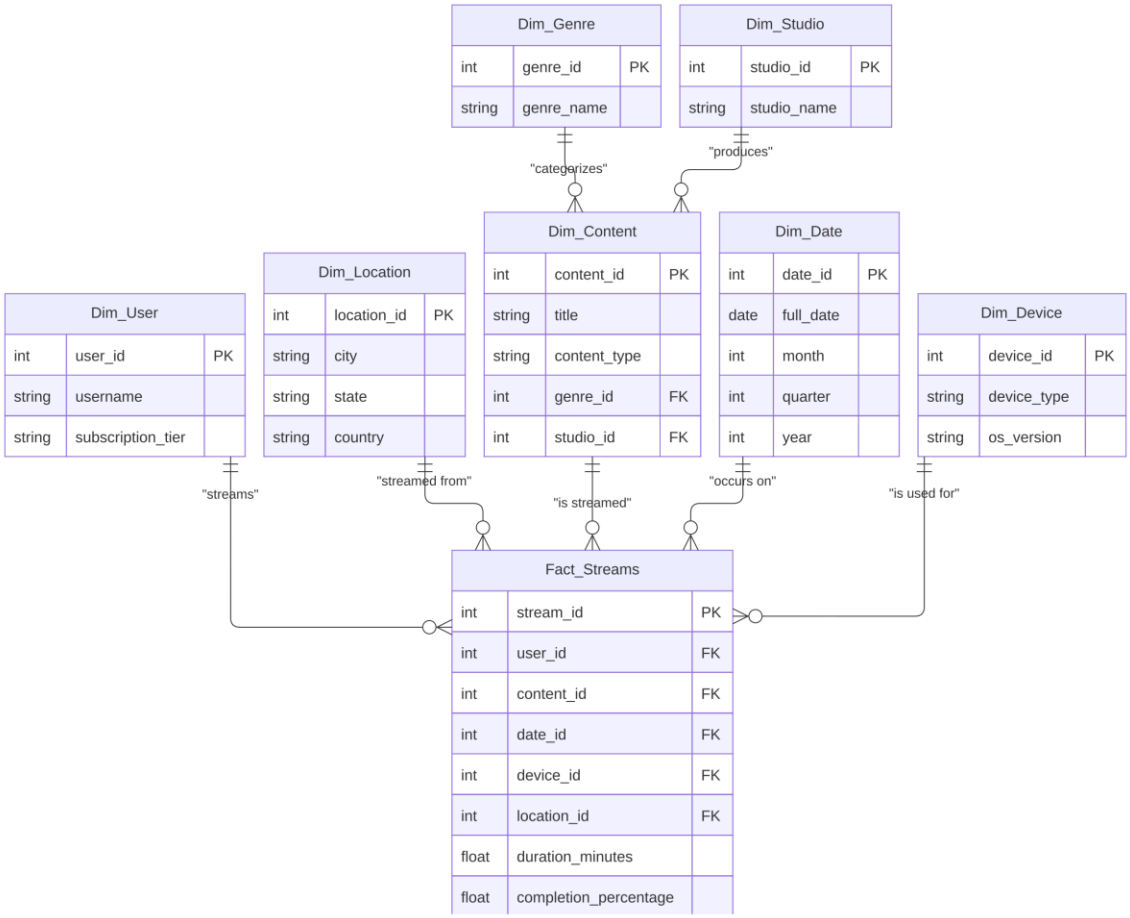
Dim_Device

Column	Description
device_id	Primary Key
device_type	Mobile / Smart TV / Laptop / Tablet
os_version	Operating system version of the device

2 . Snowflake Schema

A Snowflake Schema is a data warehouse schema design where dimension tables are normalized into smaller, separate sub-dimension tables. It extends the Star Schema structure to reduce data redundancy and save storage space by organizing attributes into hierarchical relationships.

Structure:



Main Difference :

Star schema uses a central fact table directly linked to fully denormalized dimension tables, forming a star-like shape. All dimensional data is stored in a single table, which means fewer joins and faster query performance, but at the cost of data redundancy and higher storage usage.

Snowflake schema is an extension of the star schema where dimension tables are further normalized into sub-dimension tables. For example, instead of storing genre and studio details directly inside the Content dimension, each is broken out into its own Dim_Genre and Dim_Studio table linked hierarchically. This reduces redundancy and saves storage, but requires more joins, making queries more complex and relatively slower.

Result :

Thus, the Star Schemas and Snowflake Schema for the Video Streaming Platform were successfully designed by identifying the Streaming Activity & Revenue Fact Tables and their corresponding Dimension Tables.